No invented numbers

How the design stops the AI from making up quantities, and how to check a teammate's work for it.

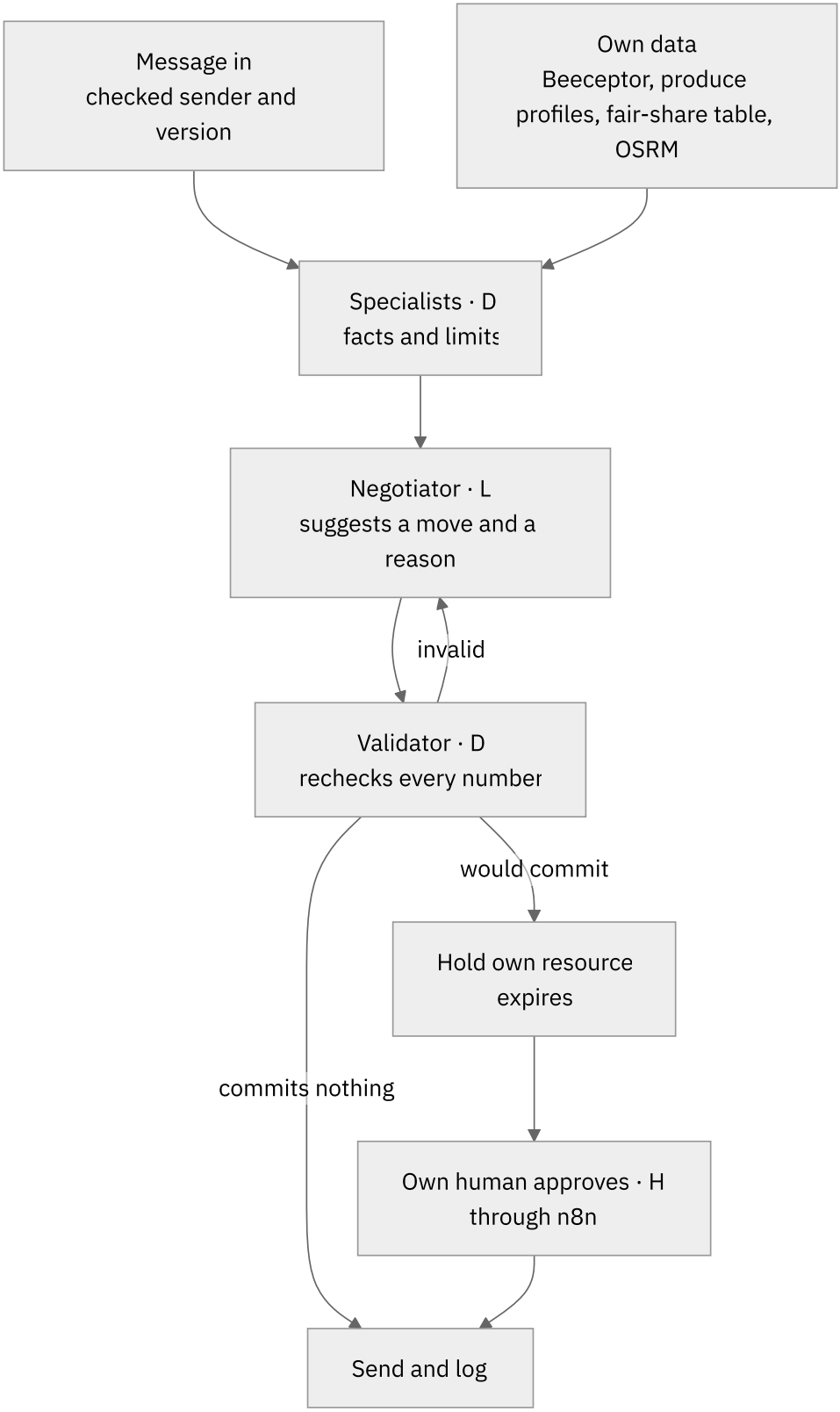
> **AFTER THIS LESSON**
> You can explain the five number rules and spot a violation in a review.

Why it matters

An agent that invents "we have room for 300 lb" can commit a food bank to something it can't do. It also makes every result meaningless. So numbers come only from calculators and data, never from the AI (ADR 0002).

The five rules

- **Calculators run first.** Every number a decision needs, such as storage space, fair-share balances, and days left, is computed before the AI is called and handed to it as a labeled fact.
- **Any extra number comes from a calculator tool.** If the AI needs, say, the pounds left after a split, it calls a tool with a strict schema. It never does arithmetic in its own text.
- **Every number in a message gets checked.** Each number in an outgoing message must match a calculator result for the current version. If one doesn't, the message isn't sent.
- **Fallback after two failures.** The agent sends a template sentence built only from calculator results, and the failure is logged.
- **Failures are counted.** Number-check failures show up as a metric in evaluation. That's honest evidence for judges.



Where the number check sits: the validator rechecks every number before anything leaves the agent.
From docs/MVP_DESIGN.md.

What the AI is for

Reading messy input (a voice call or a typed note), suggesting moves, and explaining them in plain words. A rules-based negotiator uses the same calculators, which is what makes the AI-versus-rules comparison fair.

Red flags in a teammate's code

- A prompt that asks the model to "work out" quantities, portions, or times.
- A model's text sent to another agent without the number check.
- Fair-share portions computed anywhere other than the calculator.

Source: ADR 0002, including its Consequences section.

Read next: ADR 0002, the whole file (it's short).

Previous: Lesson 8: Where Levels 1 and 2 show up · **Next:** Lesson 10: Proving it works

Quick lookup: HarvestNet at a glance.

Anything unclear? Ask me. I'm your teacher for this, and I can re-explain any part in different words.

Source of truth: docs/MVP_DESIGN.md, CONTEXT.md, and docs/adr/. If this lesson and those disagree, they win.